2D Top Down Advanced Shooter Full Game Example Template

Overview

This template is a complete, ready-to-expand 2D top-down shooter framework built in Unity. It features player combat systems (firearms, grenades, melee), reactive enemies (zombies and armed humans), detailed pickup mechanics, dynamic audio and VFX feedback, and sprite-based visual polish including outlines and shadows. All core systems are modular and designed for customization or expansion.

This project is ideal for developers wanting to build a polished top-down shooter, prototype quickly, or learn how advanced systems interact within Unity.

Please note

This documentation will guide you through the largest and most vital scripts in the template. There are more than 30 scripts all in all, so I will only cover the vital ones.

Here is a list of all of the scripts :

EnemyHealthManager.cs
EnemyZombieAIManager.cs
GameManager.cs
HumanEnemyAIManager.cs
HumanEnemyWeaponManager.cs
PickupManager.cs
PlayerAudioManager.cs
PlayerCameraManager.cs
PlayerCombatManager.cs
PlayerGrenadeManager.cs

PlayerHealthManager.cs
PlayerMeleeAttackBehaviour.cs
PlayerMovementManager.cs
PickupIdentifier.cs
AvoidEnemiesBehaviour.cs
ClampUIToCursorBehaviour.cs
DetachChildrenBehaviour.cs
EnemyProjectileBehaviour.cs
ExplosionBehaviour.cs
GrenadeBehaviour.cs
ProjectileBehaviour.cs
RandomAudioBehaviour.cs
RandomForceBehaviour.cs
RandomRotationBehaviour.cs
ScreenShakeManager.cs
ScreenshakeTrigger.cs
SpriteFadeBehaviour.cs
TimedDestroyBehaviour.cs
SpriteOutlineController.cs
SpriteShadowController.cs

Now, lets get started:

Getting Started / Setup

1. Unity Version: Use Unity 2021 LTS or higher for full compatibility.

2. Project Import: Import the template into your Unity project via the Unity Asset Store or via the provided package.

3. Scene Setup: Load the main scene from the Scenes folder.

4. Input System: Supports Unity Input Manager by default. Input mappings are editable under Project Settings > Input.

5. Tag & Layer Setup: Ensure required tags ("Player", "Enemy", etc.) and layers are set (This will be done by deafult).

6. Player Setup: The player prefab is placed in the scene. Contains all required components for gameplay.

7. Enemy Setup: Enemy prefabs (Zombie, Human) are placed manually.

8. Testing: Press Play to test core systems. Use provided weapons and pickups to explore functionality.

Main Game Flow & Architecture

- Central control is managed through the GameManager, which initializes gameplay, handles global references, and manages the state of the player and enemies as well as menu's.
- Enemy placements are managed independently and are tied to their AI behaviors.
- Player input is separated into clean managers for combat, movement, and utilities.
- Projectiles, effects, and pickups are all cleanly instantiated and cleaned up via code.

Player Systems

Movement — PlayerMovementManager

- Handles directional movement input.
- Integrates animations and optional rotation logic.
- Supports fluid, responsive top-down movement.

Combat — PlayerCombatManager

- Manages shooting logic, reloading, ammo handling, and multiple weapon types.
- Coordinates bullet instantiation and visual/audio feedback.
- Supports both hitscan and projectile-style weapons.

Health — PlayerHealthManager

- Handles taking damage, death and health and armor pickups.
- Triggers visual/audio responses on damage and death.

Grenades — PlayerGrenadeManager

- Supports throwing physics-based grenades.
- Spawns grenade objects with proper arming and explosion logic.

Melee Attacks — PlayerMeleeAttackBehaviour

- Close-range attack system with hit detection.
- Can trigger knockback, damage and death.

Audio — PlayerAudioManager

- Plays sounds based on player events: weapons, attacks, damage.

Camera Control — PlayerCameraManager

- Smooth camera following and camera distance looking.
- Works with screen shake and tracking effects.

Enemy Systems

Zombie AI — EnemyZombieAIManager

- Simple chase-and-attack AI for zombie-type enemies with some more advanced behaviours like trying to get around obstacles.
- Includes attacks, animations, pursuit logic, and damage response as well as crowd alerting.

Human AI — HumanEnemyAIManager

- Advanced AI with vision, cover-seeking, strafing and projectile combat.
- Supports attacking logic.
- Includes attacks, animations, pursuit logic, and damage response as well as crowd alerting.

Enemy Weapons — HumanEnemyWeaponManager

- Handles AI shooting logic, bullet control, reloading, audio and cooldowns.
- Works similarly to the player weapon system. Derived from the same codebase.

Enemy Health — EnemyHealthManager

- Same principles as player health but optimized for AI logic.
- Includes damage reaction, death events and more.

Enemy Projectiles — EnemyProjectileBehaviour

- Defines projectile motion, collision, and damage handling.

- Visual differentiation between player and enemy projectiles.
- Creates reaction feedback and hit visuals on collision.

Weapon System

Multi-Gun Support

- Each weapon has its own settings: damage, fire rate, reload time, bullet spread, ammo, sounds, sprite and more.
- Can easily add new weapons by defining a config and assigning visuals.

Shooting & Reloading

- Shooting instantiates bullets with weapon-specific properties.
- Audio cues, muzzle flashes, and animation triggers included.

Integration

- Weapons are shared across player/enemy systems through consistent projectile and damage handling logic.

Pickup System

PickupManager & PickupIdentifier

- Controls pickup spawning and logic (health, ammo, grenades, new guns).
- Pickups can enable/disable based on player state.
- Identifiers define pickup type and logic to apply on player contact.

Audio System

Clip Structure

- Organized by action type: fire, reload, damage.
- Supports multiple variations for sound randomization.

Playback Logic

- Rate-limited to prevent spam.
- Used across players, enemies, and environment objects.

Visual & Effects Systems

SpriteOutlineController

- Adds dynamic outlines around characters or objects.
- Helpful for clarity in dark or busy scenes.
- Used to identify pickups and more.

SpriteShadowController

- Adds simple but effective real-time controllable shadows under characters.
- Gives a sense of depth to flat sprites.

ScreenShakeManager & ScreenshakeTrigger

- Used for gunfire, explosions, and melee impact.
- Shakes are customizable in duration, strength, and delays.

SpriteFadeBehaviour

- Fades out sprites over time (e.g., dead enemies, explosions, blood).
- Used to clean up visuals after death or destruction.

Decals & Projectile Effects

- Bullet impacts spawn decals (blood, sparks, dirt).
- Each gun can have different decal types and spawn rules.

Bullet Sprites per Weapon

- Weapons define their own projectile visuals.
- Bullets can vary in size, speed, color, and trail style.

Player Projectile System

- Central system for spawning projectiles tied to weapon settings.
- Modular for possible future extension (e.g., fire bullets, bouncing, AoE).

Bullet Feedback & Screen Shake

- Guns can trigger different feedback effects.
- Guns always trigger hit feedback effects.
- Feedback includes camera shake, hit decals, sounds, and animations.

Utility & Helper Scripts

RandomAudioBehaviour

- Plays random clips from a list.
- Used for varied footsteps, pain sounds, etc.

RandomForceBehaviour / RandomRotationBehaviour

- Adds physics variation to spawned objects.
- Used for debris, pickups, or enemy death effects.

TimedDestroyBehaviour

- Automatically destroys objects after delay.
- Used for projectiles, decals, and effects.

DetachChildrenBehaviour

- Detaches child objects from their parent.
- Useful for preserving effects after object death.

ClampUIToCursorBehaviour

- Keeps UI elements aligned with the mouse or gamepad cursor.
- Used for the crosshair in this case.

AvoidEnemiesBehaviour

- Used by objects that should not overlap with enemies (e.g., pickups, and other enemies themselves).
- Applies logic to reposition or repel on spawn.

Outro

This template offers a solid and advanced foundation for creating a complete 2D top-down shooter game. Whether you're developing a commercial title, a game jam prototype, or a personal project, the modular structure and extensible systems give you the flexibility to iterate quickly and scale efficiently.

Use the included components as-is or customize them to match your vision. With weapon variety, reactive enemies, polished visuals, and a complete gameplay loop, you're equipped to build something unique.

Happy developing!

Contact information

For any queries or questions as well as suggestions please reach me at Johannesnienaber@gmail.com for email or through the Aligned Games studio website here https://alignedgames.mailchimpsites.com/

Thank you for purchasing the Template package, I hope you enjoy using it and make an awesome game with it!